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validation engineers with a set of requirements to complete the verification & validation (V&V) process, and then report back on the results. However, in the concurrent design and test approach, there are real-time testing tasks going on throughout the entire process. "In the design phase, engineers perform model-in-the-loop testing by applying stimulus profiles or test scripts to simulations of the system and analysing its response. Once an algorithm is ready for prototyping, engineers take the prototype controller to the test lab or field test site for evaluation after which it is deployed. Testing occurs throughout the process," adds Chivukula.

T. Anand of Knewron also stresses on the importance of testing during design, and cites an example where it helped his firm to remedy a problem quite inexpensively. "In one MCU-based design, we found that RTC value was not synchronised; rather it was giving random timing whenever invoked. Majority of the system functionality was dependent on RTC sync and so was a critical issue. By logic everything should have worked fine, but it didn't. We could get out from that issue after changing declarations of RTC-related variables and also changing their types. Additionally, we had to change compiler/linker settings where it was optimising unnecessarily and causing trouble. If we would have skipped that test (assuming that RTC code is common and can't have any issue) we would have spent too much time in final phases during release testing, which is costlier than early detection and remedy."

Advantages to the design process

As mentioned earlier, in many cases, it is not possible to complete a design without testing during the design phase—especially for new technology. Apart from that, the other major advantages are risk reduction and cutting costs. If a design has a problem that is only discovered after the prototype is almost complete, the costs of correcting it can be enormous. Stephen Hire explains, "If the problem is at the heart

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of the design, then a lot of rework will be required; and where companies are pushing to meet a specific launch date, a problem found late in the design phase can cause big delays. Or even worse, the problems may only be found after the launch when customers start complaining and returning products to the manufacturer—every manufacturer and designer needs to minimise the risk of this happening."

Teledyne Lecroy spokesman gives us a different point of view, "Product development is fundamentally a risk mitigation exercise, and testing gives you confidence that the risks have been mitigated. If a given subsystem has been thoroughly tested and verified, it should not be the source of issues upon assembly and testing of the entire system. If all of the subsystems have been similarly verified, faults in the end product can typically be traced to issues with the interfaces between the subsystems. Particularly with larger systems, testing during the design phase helps to partition the problems to achieve faster time to market."

In the design of any large, complex system, testing during design implies that you know where you are in the process. Knowing where you are as you design can inform what you do at the next step of the design process. In general, if implemented properly, testing during design reduces, and has the potential for eliminating, long loops of